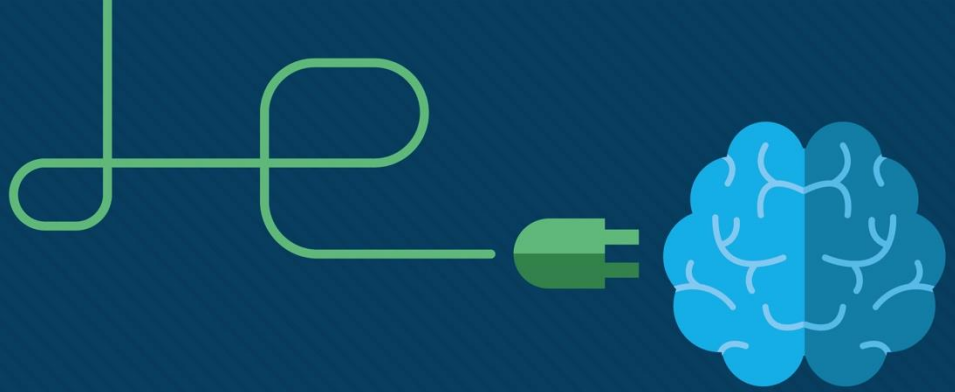

Module 4
First Hop Redundancy

ITNET03

Advanced Routing and Switching

Module Objectives

Module Title: First Hop Redundancy

Module Objectives:

- Explain the purpose and operation of first hop redundancy protocols.
- Explain how the Hot Standby Router Protocol operates
- Implement first hop redundancy in a network using HSRP

Module References:

- CCNAv7 SRWE – Module 9

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4.1 First Hop Redundancy

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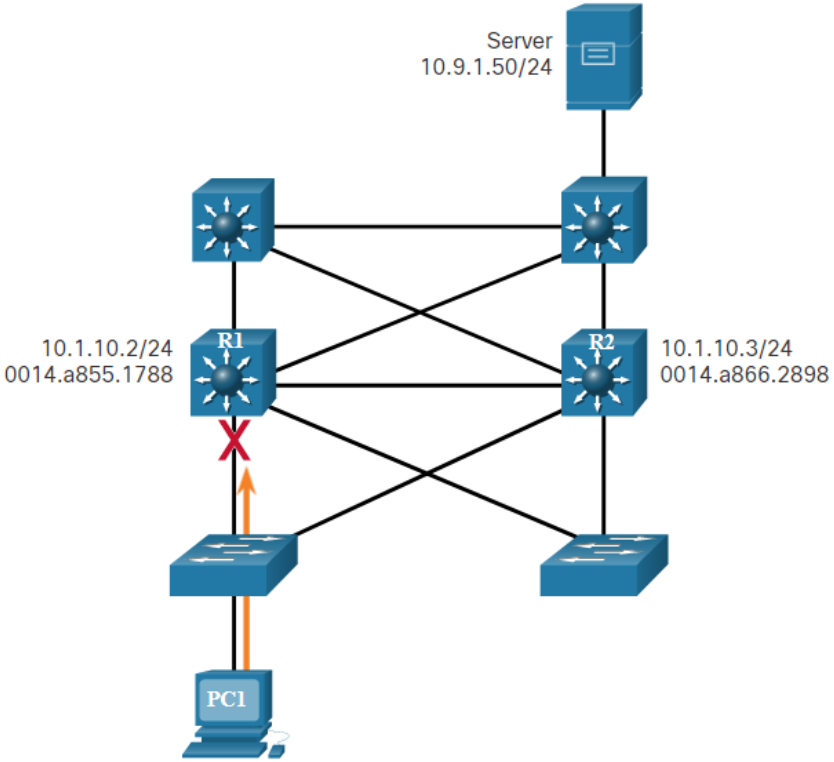
Default Gateway Limitations

- End devices are typically configured with a single default gateway IPv4 address.
  - If the default gateway router interface fails, LAN hosts lose outside LAN connectivity.
  - Even if a redundant router or Layer 3 switch that could serve as gateway exists, network hosts cannot automatically use it
- A mechanism is needed to provide alternate default gateways in switched networks where two or more routers are connected to the same network segment



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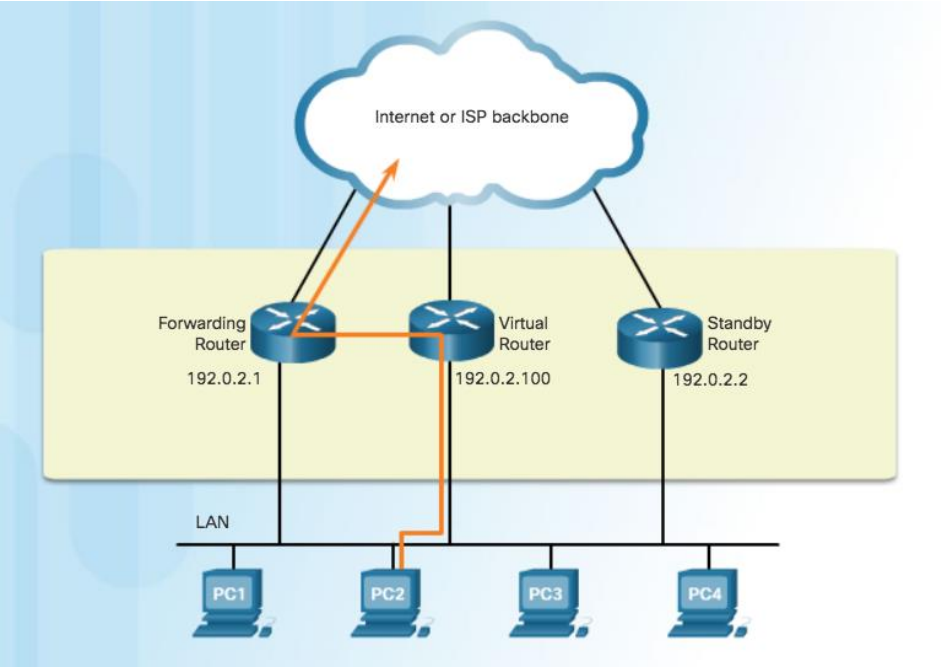
Router Redundancy

- To prevent a single point of failure at the default gateway, implement a virtual router.
  - Multiple physical routers present the illusion of a single router to the hosts on the LAN.
  - Physical router that performs actual traffic forwarding is transparent to the host devices.
  - In case the forwarding router fails, any of the remaining physical routers can take over the forwarding task automatically
  - Ability of a network to dynamically recover from the failure of a device acting as a default gateway is known as **first-hop redundancy**.



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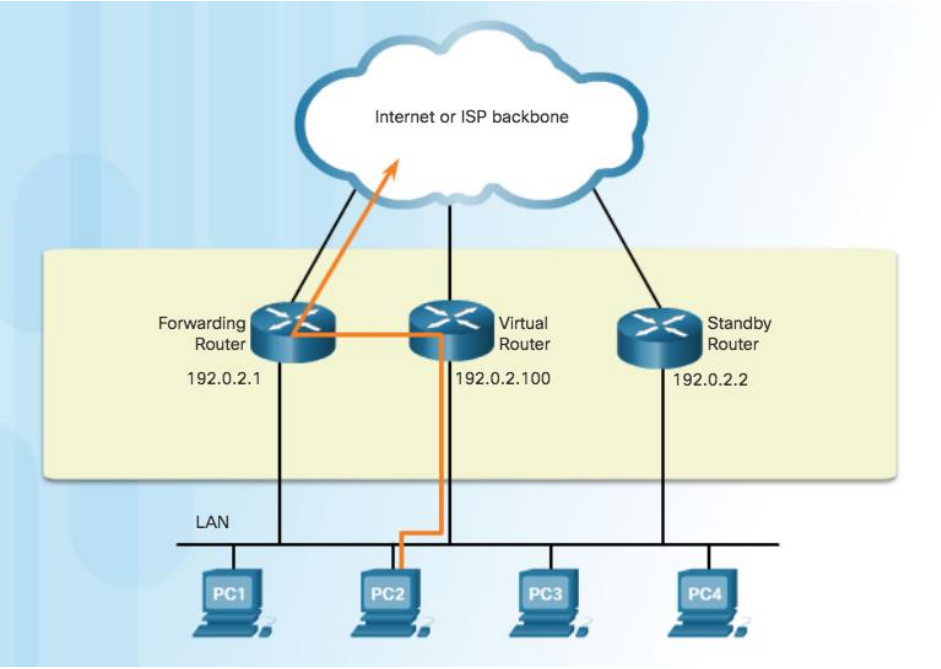
Router Redundancy

- Two or more routers can act as a single virtual router by sharing an IP address and a MAC address
- A redundancy protocol provides the mechanism for determining which router should take the active role in forwarding traffic.
- IPv4 address of the virtual router is configured as the default gateway for the workstations on a specific IPv4 segment.
- Frames that are sent to the virtual router are physically processed by the currently active router within the virtual router group.



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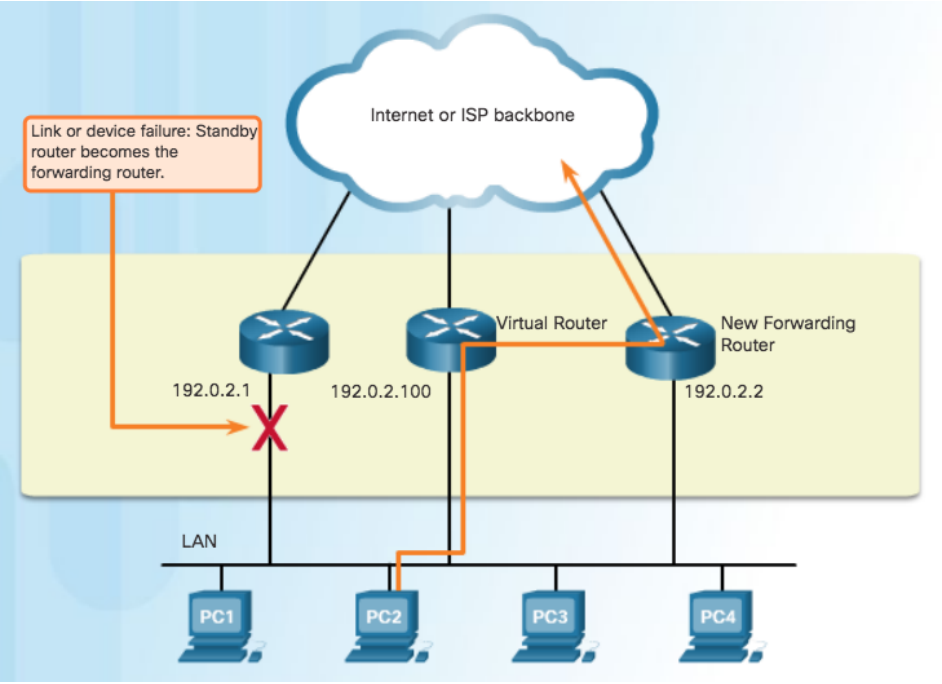
Steps for Router Failover

- When the active router fails, the redundancy protocol transitions the standby router to the new active router role.
- These are the steps that take place when the active router fails:

1. The standby router stops seeing periodic “heartbeat” messages from the forwarding router.
2. The standby router assumes the role of the forwarding router.
3. Because the new forwarding router assumes the identity of the forwarding router, the host devices see no disruption in service.



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First Hop Redundancy Protocols

- **Hot Standby Router Protocol (HSRP)**
  - A Cisco-proprietary FHRP designed to allow for transparent failover of a first-hop IPv4 device.
  - Active device is the device that is used for routing packets and standby device is the device that takes over when the active device fails.
- **HSRP for IPv6**
  - Cisco-proprietary FHRP providing the same functionality of HSRP, but in an IPv6 environment.
  - Uses periodic router advertisements (RAs) to advertise the HSRP virtual IPv6 link-local address

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First Hop Redundancy Protocols (Cont.)

- **Virtual Router Redundancy Protocol version 2**
  - A nonproprietary protocol that dynamically assigns responsibility for one or more virtual routers to the VRRP routers on an IPv4 LAN.
  - One router is elected as the virtual router master, with the other routers acting as backups, in case the virtual router master fails.
- **VRRPv3** Capability to support IPv4 and IPv6.
  - Updated version of VRRP that supports both IPv4 and IPv6.
- **Gateway Load Balancing Protocol (GLBP)**
  - Cisco-proprietary FHRP that protects data traffic from a failed router or circuit allowing **load balancing** between a group of redundant routers.
- **GLBP for IPv6 -** Cisco-proprietary FHRP providing the same functionality of GLBP.
  - Cisco-proprietary FHRP providing the same functionality of GLBP for IPv6 networks

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4.2 Hot Standby Router Protocol

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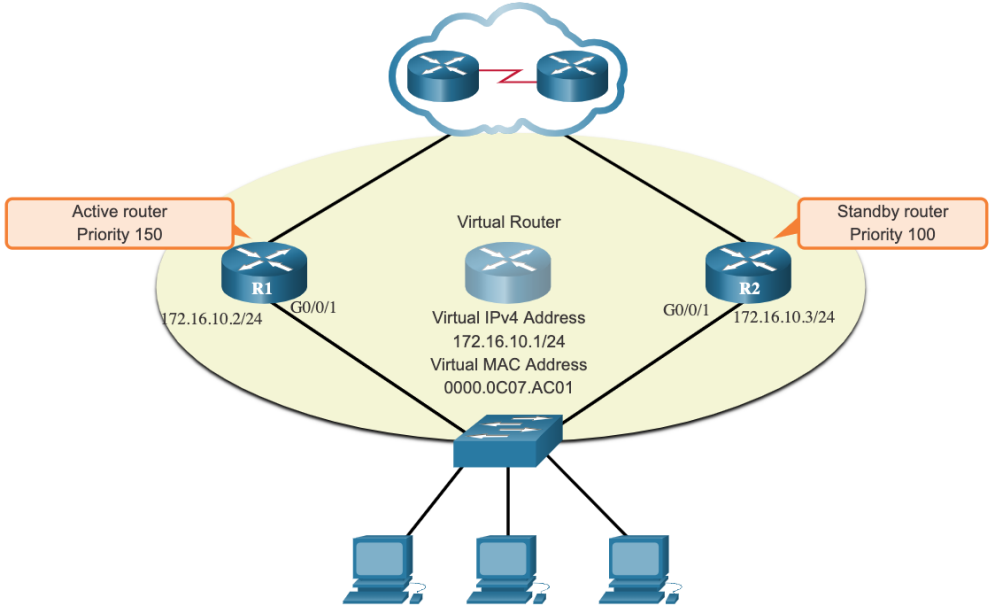
HSRP Overview

- HSRP and HSRP for IPv6 are Cisco-proprietary FHRP that are designed to allow for transparent failover of a first-hop IP device to avoid losing outside network access if the default router fails.
- Ensures high network availability by providing first-hop routing redundancy for IP hosts on networks configured with an IP default gateway address.
- Used in a group of routers for selecting an active device which routes packets, and a standby device which monitors the status of the active device and takes over packet forwarding responsibilities when the active device fails



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HSRP Router Roles

- **Virtual** (1 per group)
  - IP and MAC address pair configured as default gateway of end devices.
  - NOT a physical device – only a representation of the network gateway
- **Active** (1 per group)
  - Sends periodic hello messages to the HSRP group multicast address
  - Physical router that performs actual processing to route packets sent to the virtual router MAC address
- **Standby** (1 per group)
  - Sends periodic hello messages to the HSRP group multicast address
  - Takes over as active router if hellos stop arriving from the active router
- **Others**
  - Routers that are not serving as active / standby but are still members of the group
  - Will contend for active and standby roles if either router fails

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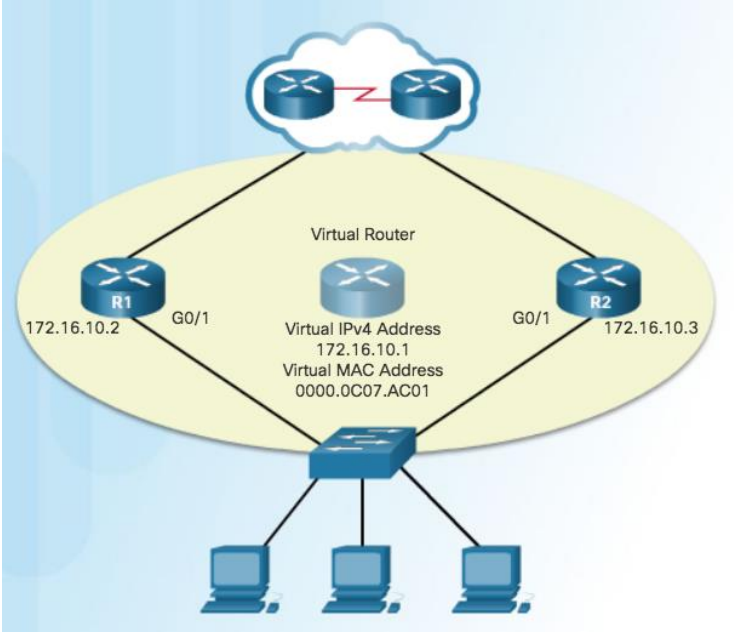
HSRP Versions



Version	HSRP V1 (Default)	HSRP V2
Group numbers	0 to 255	0 to 4095
Multicast address	224.0.0.2	224.0.0.102 or FF02::66
Virtual MAC address	0000.0C07.AC00 - 0000.0C07.ACFF (last two digits group number)	**IPv4** 0000.0C9F.F000 to 0000.0C9F.FFFF **IPv6** 0005.73A0.0000-0005.73A0.0FFF (last three digits of group number)
Support for MD5 authentication	No	Yes

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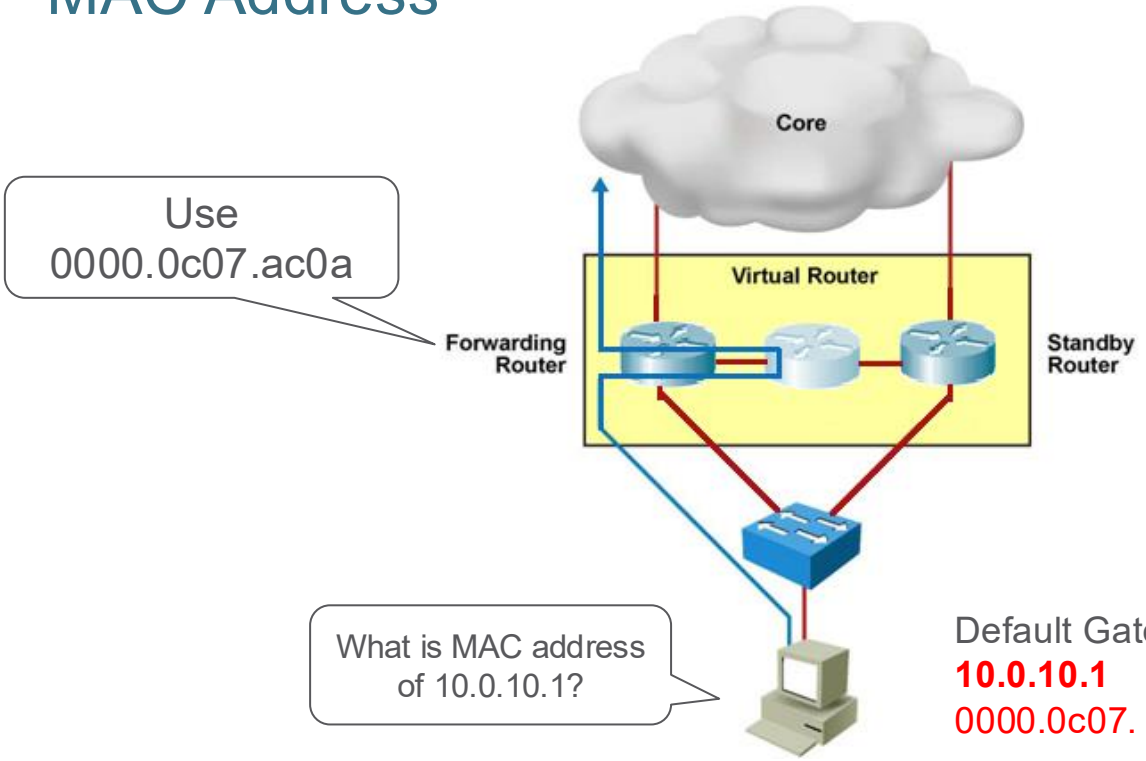
HSRP MAC Address



Router A: 10.0.10.2
Router B: 10.0.10.3
Virtual Router: 10.0.10.1

0000.0c07.ac0a

Default Gateway:
10.0.10.1
0000.0c07. ac0a

- HSRP MAC address: **0000.0c07.acxx**, where xx is the HSRP group identifier

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HSRP States and Timers

Initial
- Starting state entered via a configuration change or interface enable – HSRP not yet running

Learn
- The router has not received a hello message from the active router yet does not know the virtual IP address yet

Listen
- Router is neither the standby or active router yet but knows virtual IP address and listens for hellos

Speak
- Router knows virtual IP address and sends hellos to participates in standby and active router election

Standby
- Router sends hellos and is candidate to be next active router

Active
- Router sends hellos and currently forwards packets that are sent to the group virtual MAC address

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HSRP States and Timers

- Timers influence the speed at which the network recovers from a failed gateway
- **Hello Time**
  - Interval between periodic hello messages from the active and standby routers
  - Default = 3 secs Range = 1 to 255 secs
- **Hold Time**
  - Time before the active or standby router is declared to be down if no hello is received
  - Default = 10 secs Range = 1 to 255 secs
  - Ideally at least 3x the length of hello time
- Value for timers may be set independently per HSRP group
  - Lower interval = faster response to router failure but higher demand on resources
  - Higher interval = slower response to router failure but lower demand on resourced

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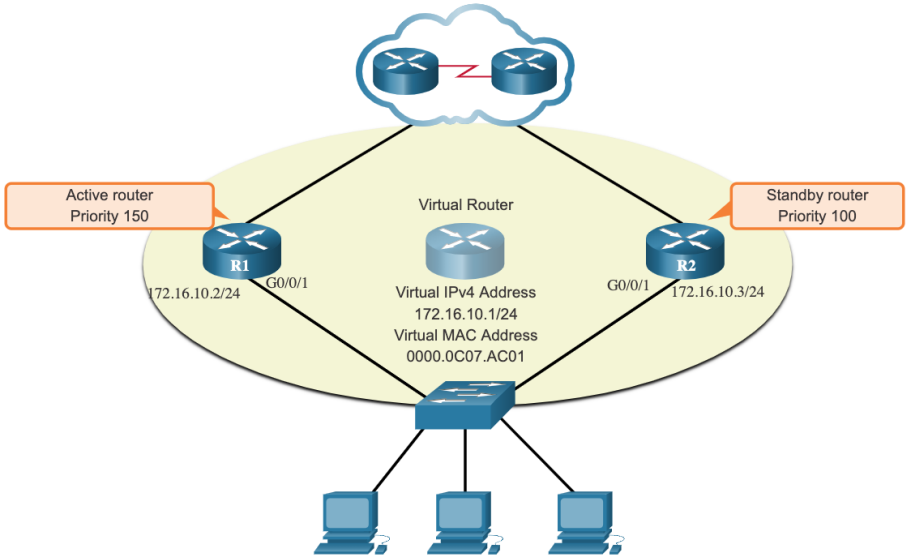
HSRP Priority and Preemption

- The role of the active and standby routers is determined though an election process using the HSRP **priority** as basis.
  - The router with the **highest HSRP priority** within a group will become the active router.
  - The router with 2nd highest priority becomes the standby router
- If router priorities within a group are equal, the router with the numerically highest IPv4 address is elected the active router.



- By default, after a router becomes the active / standby router, it will remain in its role if another router comes online with a higher HSRP priority
- **Preemption** is the optional ability of an HSRP router to trigger the re-election process when it comes online to assume the active router role when it has a higher priority than the current active router in the HSRP group

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4.3 HSRP Configuration

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HSRP Configuration Procedure

Task	IOS Command
Set IP address and subnet mask of an interface/subinterface	**Router(config)# interface** *int_id* **Router(config)# ip address** *address subnet_mask*
Set HSRP version (optional, default 1)	**Router(config-if)#standby version** ***1|2***
Configure HSRP group and virtual IP address on the interface *1 group, 1 virtual address	**Router(config-if)#standby** *grp_no* **ip** *virtual_address*
Set router HSRP priority (optional but recommended, default 100) *value range: 0-255	**Router(config-if)#standby** *grp_no* **priority** *value*
Enable preemption (optional but recommended)	**Router(config-if)#standby** *grp_no* **preempt**
Set HSRP group timers (optional) *make sure ALL routers in the group match timers	**Router(config-if)#standby** *grp_no* **timers** *hello_interval hold_time*
Enable interface tracking (optional)	**Router(config-if)#standby** *grp_no* **track** *interface_id*

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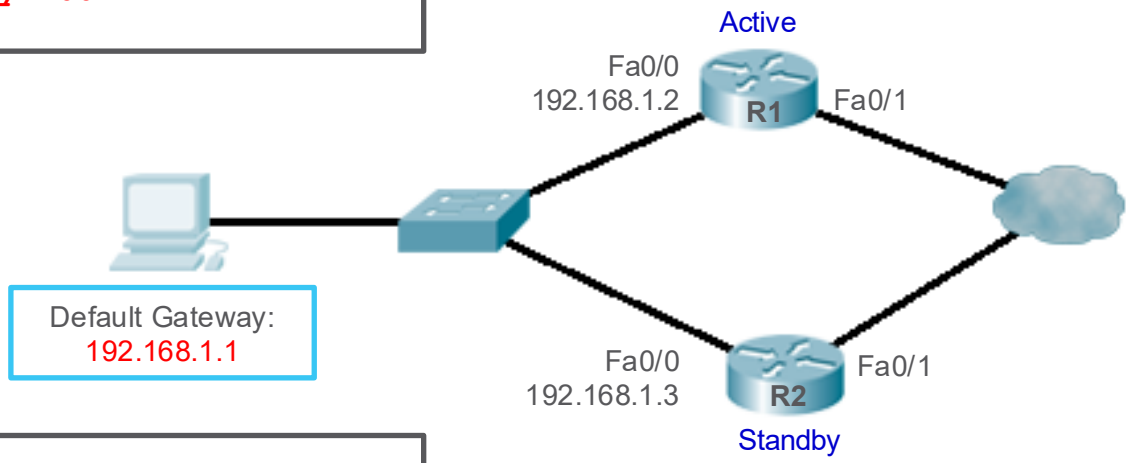
Sample HSRP Configuration

```
R1(config)#interface Fa0/0
R1(config-if)#ip address 192.168.1.2 255.255.255.0
R1(config-if)#standby 1 ip 192.168.1.1
R1(config-if)#standby 1 priority 100
R1(config-if)#standby 1 preempt
```




```
R2(config)#interface Fa0/0
R2(config-if)#ip address 192.168.1.3 255.255.255.0
R2(config-if)#standby 1 ip 192.168.1.1
R2(config-if)#standby 1 priority 95
R2(config-if)#standby 1 preempt
```

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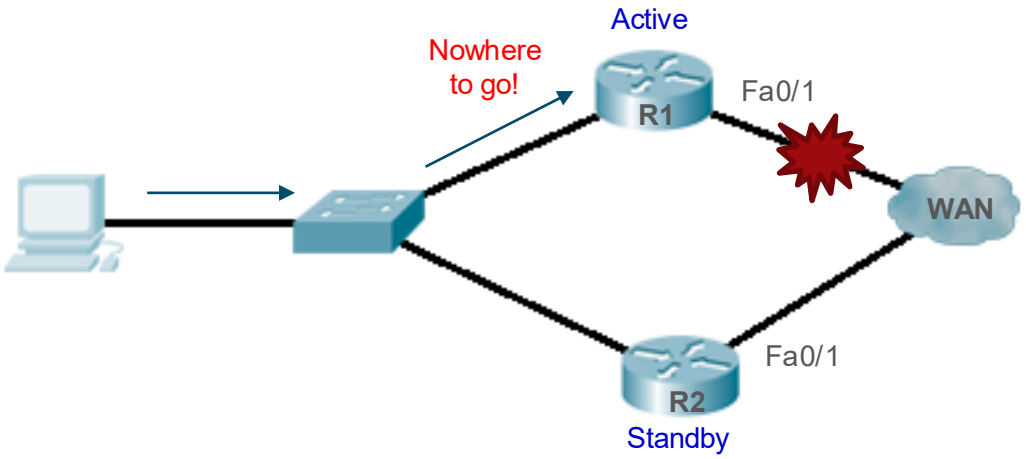
HSRP Interface Tracking

- Scenario
  - Router 1 is active router based on priority
  - Both routers connect to the WAN through their Fa0/1
  - Problem: If Fa0/1 of Router1 goes down, the host will still use Router1 as the active gateway to the WAN



- Solution: HSRP Interface Tracking - enables the priority of an HSRP group router to be automatically adjusted based on availability of a given interface
- When a tracked interface becomes unavailable, the HSRP priority of the router is automatically lowered
- Purpose is to allow a router to relinquish the active role if a key interface becomes unavailable

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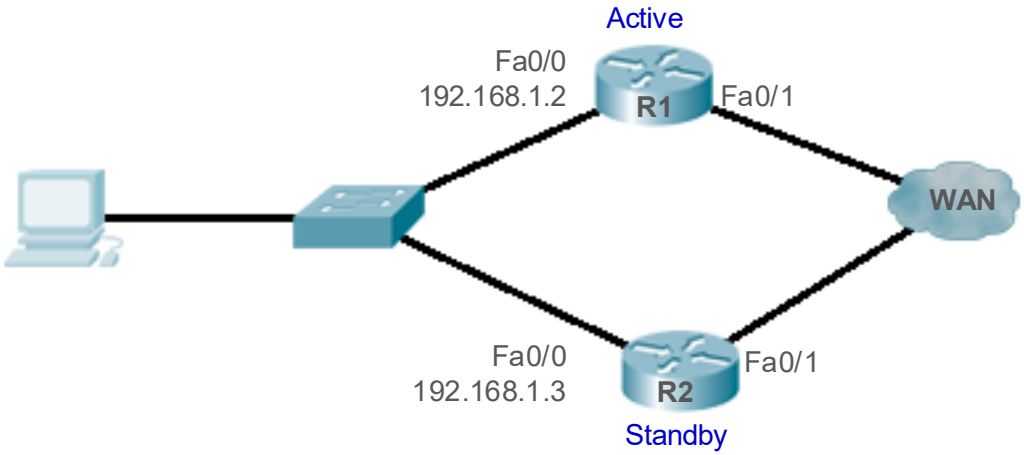
HSRP Interface Tracking

- Configuration command: `Router(config-if)#standby grp_no track int_id`
- Usually enabled in conjunction with preemption
- Default decrease in priority is by 10
- To fix the issue scenario:
  - Enable tracking on the routers for Fa0/1.
  - If Fa0/1 goes down, whichever is the active router must decrease priority to give up the active role

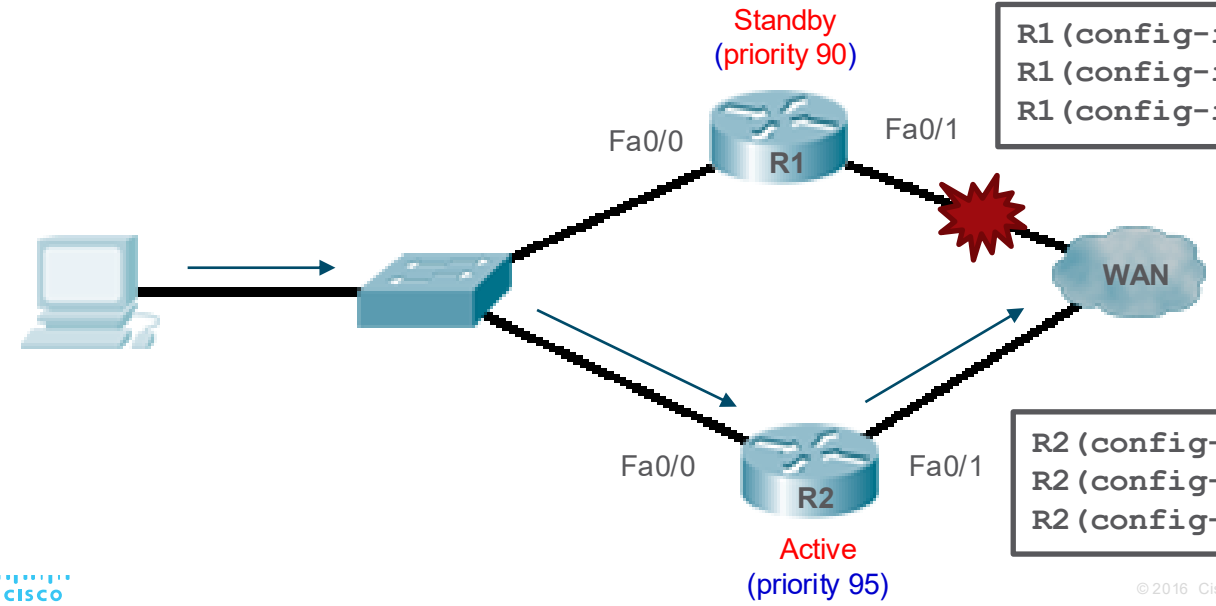
```
R1(config-if)#standby 1 priority 100
R1(config-if)#standby 1 preempt
R1(config-if)#standby 1 track Fa0/1
```




```
R2(config-if)#standby 1 priority 95
R2(config-if)#standby 1 preempt
R2(config-if)#standby 1 track Fa0/1
```

HSRP Interface Tracking

- Under normal operations, all network communications existing to the WAN go through R1
- When R1 link to ISP goes down, its priority is decreased by 10 and is now lower than R2 priority
- R2 forces reelection (preemption enabled) and takes over active role
- LAN traffic flows through R2 to the WAN




```
R1(config-if)#standby 1 priority 100
R1(config-if)#standby 1 preempt
R1(config-if)#standby 1 track Fa0/1
```

```
R2(config-if)#standby 1 priority 95
R2(config-if)#standby 1 preempt
R2(config-if)#standby 1 track Fa0/1
```

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HSRP Verification

- show standby – Displays interfaces with detailed HSRP group membership and settings on the routers

```
R1#show standby
FastEthernet0/0 - Group 1
  State is Active
    6 state changes, last state change 00:10:51
  Virtual IP address is 192.168.1.1
  Active virtual MAC address is 0000.0C07.AC01
    Local virtual MAC address is 0000.0C07.AC01 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 1.131 secs
  Preemption enabled
  Active router is local
  Standby router is 192.168.1.3
  Priority 100 (default 100)
    Track interface FastEthernet0/1 state Up decrement 10
  Group name is hsrp-Fa0/0-1 (default)
```

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HSRP Verification

- show standby brief – Displays summary of HSRP-enabled interfaces and group settings

```
R1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Fa0/0       1    100 P Active   local           192.168.1.3     192.168.1.1
```

- show standby neighbors– Displays interface HSRP group membership and neighbors routers that are members of the same group

```
R1#show standby neighbors Fa0/0

HSRP neighbors on FastEthernet0/0
  192.168.1.3
    No Active groups
    Standby groups: 1
    BFD enabled
```

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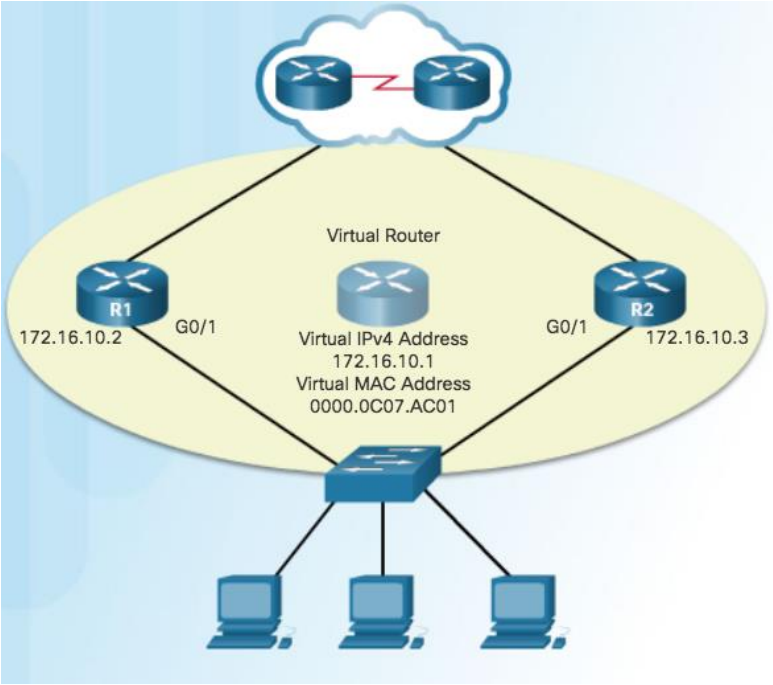
HSRP Failure

- Most issues will arise during one of the following HSRP functions:
  - Failing to successfully elect the active router that controls the virtual IP for the group
  - Failure of the standby router to successfully keep track of the active router
  - Failing to determine when control of the virtual IP for the group should be handed over to another router
  - Failure of end devices to successfully configure the virtual IP address as the default gateway



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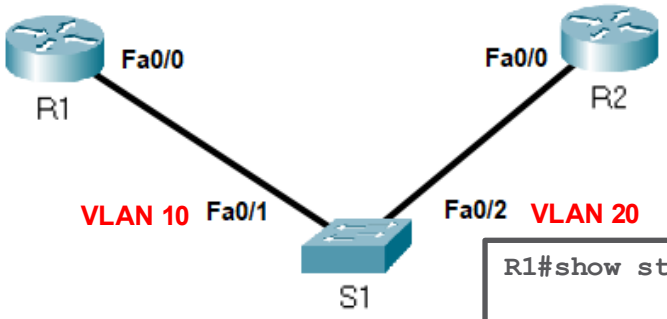
Common HSRP Configuration Issues

Case 1: HSRP routers are not connected to the same network segment.

- This can be a physical layer issue, a VLAN configuration issue or a trunking issue.



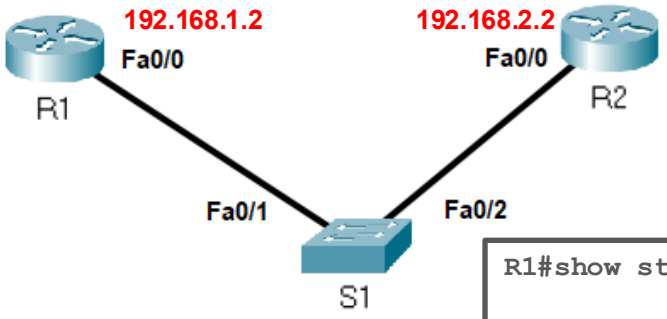

```
R1#show standby brief
                     P indicates configured to preempt.
                     |
Interface    Grp  Pri P State    Active          Standby         Virtual IP
Fa0/0        1    100 P Active   local           unknown         192.168.1.1
```

```
R2#show standby brief
                     P indicates configured to preempt.
                     |
Interface    Grp  Pri P State    Active          Standby         Virtual IP
Fa0/0        1     95 P Active   local           unknown         192.168.1.1
```

Common HSRP Configuration Issues

Case 2: HSRP routers are not configured with IPv4 addresses from the same subnet.

- HSRP hello packets are local. They are not routed beyond the network segment. Therefore, a standby router would not know when the active router fails.




```
R1#show standby brief
                     P indicates configured to preempt.
                     |
Interface    Grp  Pri P State    Active          Standby         Virtual IP
Fa0/0        1    100 P Active   local           unknown         192.168.1.1
```

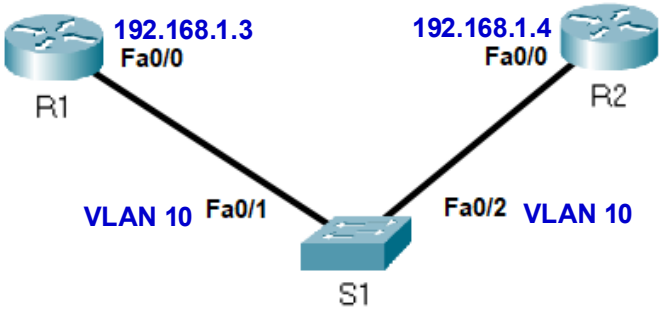
```
R2#show standby brief
                     P indicates configured to preempt.
                     |
Interface    Grp  Pri P State    Active          Standby         Virtual IP
Fa0/0        1     95 P Active   local           unknown         192.168.1.1
```

Common HSRP Configuration Issues

Case 3: HSRP routers are not configured with the same virtual IPv4 address.

```
R1(config-if)#standby 1 ip 192.168.1.1
```

```
R2(config-if)#standby 1 ip 192.168.1.2
```




```
R1#show standby brief
                     P indicates configured to preempt.
                     |
Interface    Grp  Pri P State    Active          Standby         Virtual IP
Fa0/0        1    100 P Active   local           unknown         192.168.1.1
```

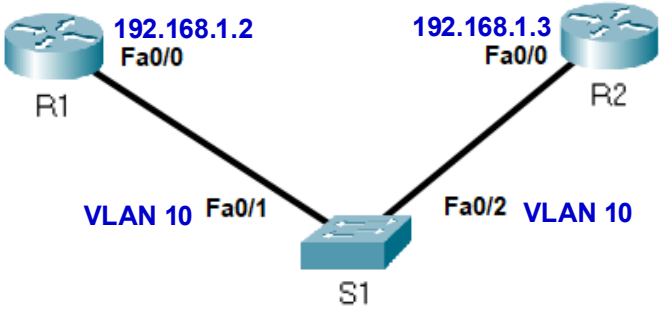
```
R2#show standby brief
                     P indicates configured to preempt.
                     |
Interface    Grp  Pri P State    Active          Standby         Virtual IP
Fa0/0        1     95 P Active   local           unknown         192.168.1.2
```

Common HSRP Configuration Issues

Case 4: HSRP routers are not configured with the same HSRP group number.

```
R1(config-if)#standby 1 ip 192.168.1.1
```

```
R2(config-if)#standby 2 ip 192.168.1.1
```




```
R1#show standby brief
                     P indicates configured to preempt.
                     |
Interface    Grp  Pri P State    Active          Standby         Virtual IP
Fa0/0        1    100 P Active   local           unknown         192.168.1.1
```

```
R2#show standby brief
                     P indicates configured to preempt.
                     |
Interface    Grp  Pri P State    Active          Standby         Virtual IP
Fa0/0        2     95 P Active   local           unknown         192.168.1.1
```

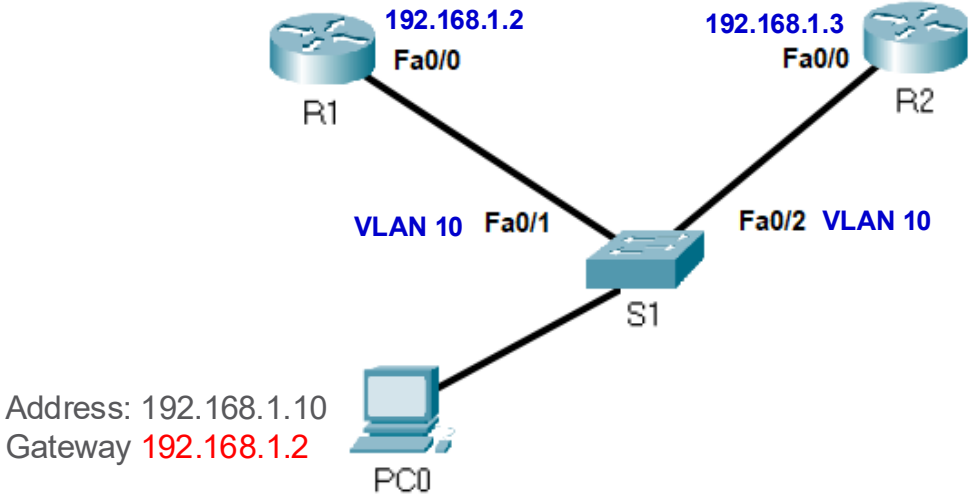
Common HSRP Configuration Issues

Case 5: End devices are not configured with the correct default gateway address.

- This will cause hosts to lose connectivity to hosts outside their own subnet, or fail to switchover to a working gateway when the active router changes in the network

```
R1(config-if)#standby 1 ip 192.168.1.1
```

```
R2(config-if)#standby 1 ip 192.168.1.1
```



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4.4 Redundancy Integration

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Using STP and FHRP Together

- Per-VLAN Spanning Tree (PVST+/Rapid PVST+) and First Hop Redundancy Protocols (HSRP, VRRP, or GLBP) should be intentionally aligned so that Layer 2 forwarding paths and Layer 3 gateway selection reinforce each other.
- For each VLAN, the switch that is the STP root bridge should also be the active/default gateway for that VLAN.
  - End-host traffic reaches the default gateway without unnecessary hops
  - Traffic does not traverse blocked STP links
  - Inter-VLAN and upstream traffic follow the shortest paths

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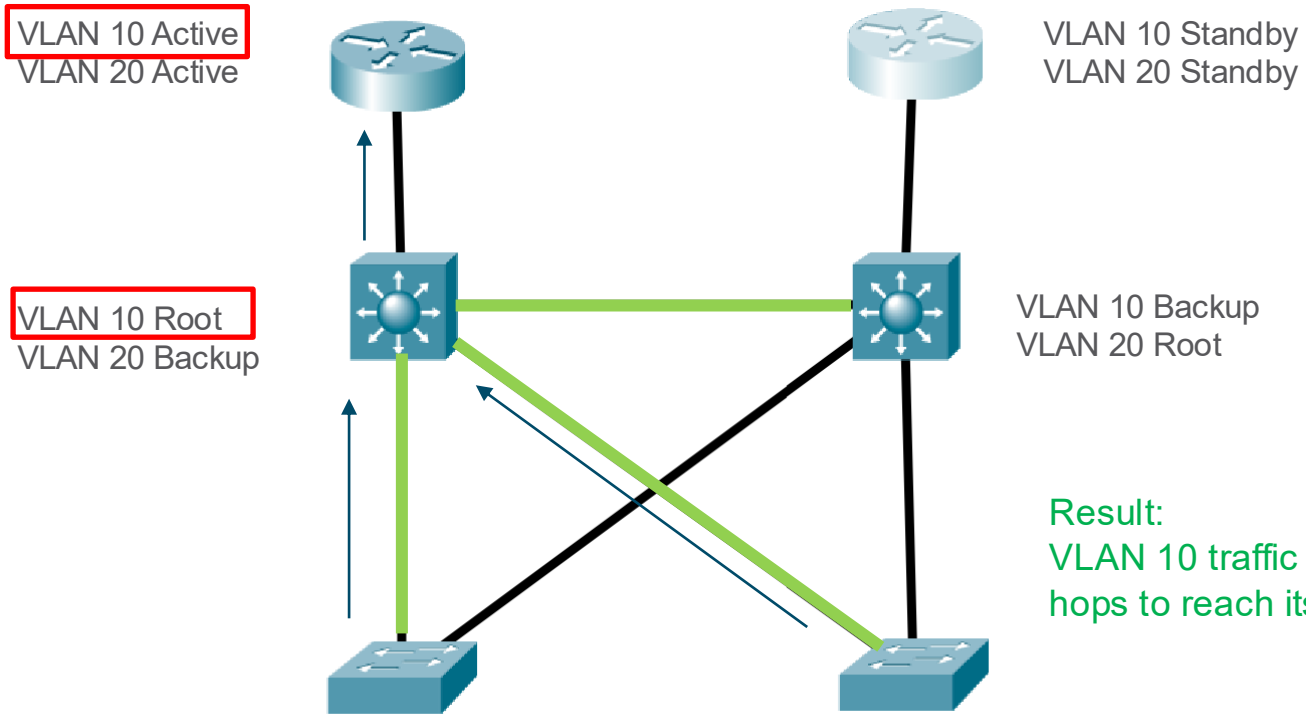
STP and FHRP Alignment



Result:
VLAN 10 traffic needs minimal hops to reach its gateway

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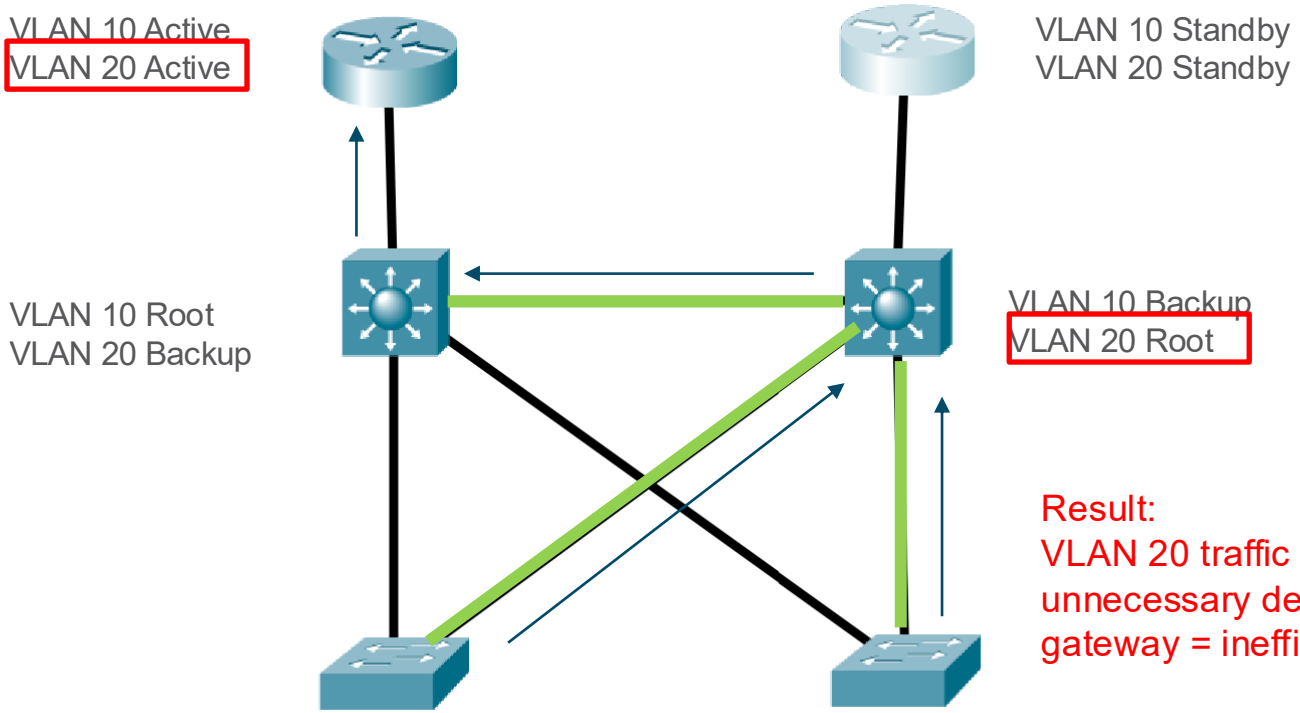
STP and FHRP Misalignment



Result:
VLAN 20 traffic takes an unnecessary detour to reach its gateway = inefficient path

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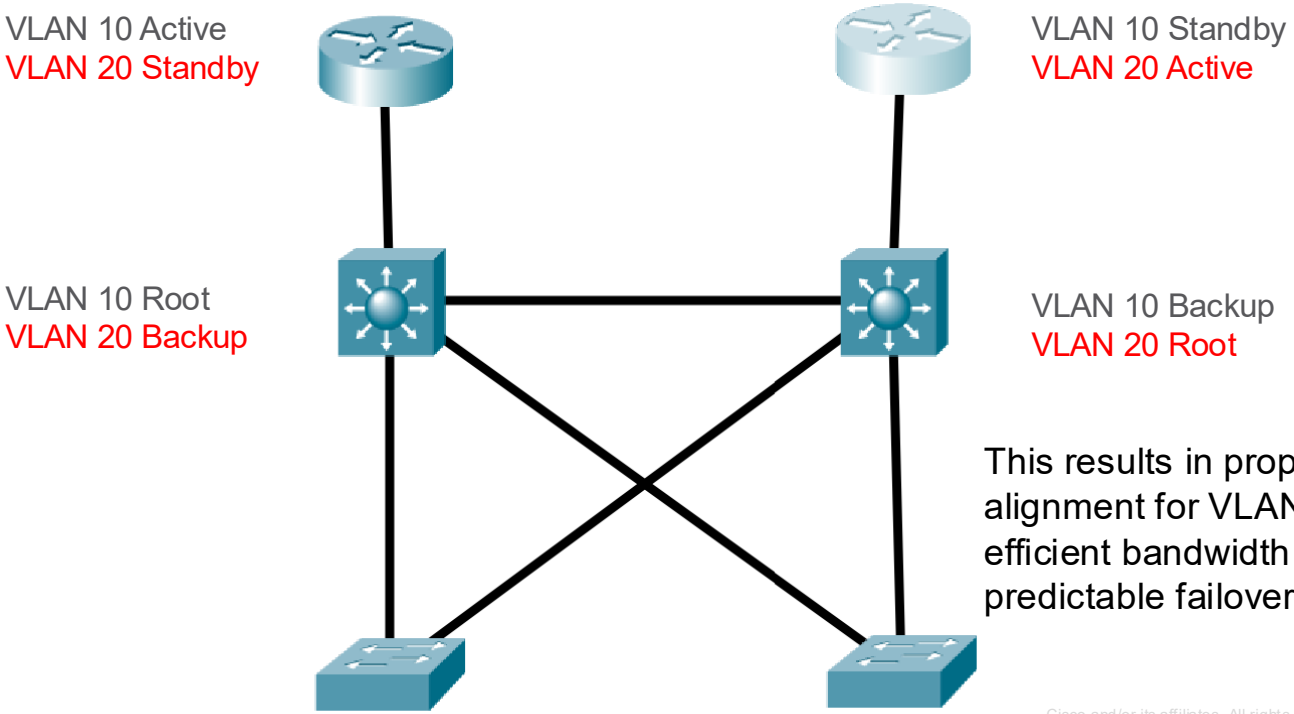
Fixing the Issue

Manipulate STP and FHRP priorities to deliberately set and align primary and backup devices



This results in proper STP and FHRP alignment for VLANs – leading to more efficient bandwidth utilization and predictable failover behavior

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What did you learn in this module?

- First Hop Redundancy provides alternate default gateways in switched networks where two or more routers are connected to the same work together to present the illusion of a single router to the hosts on the LAN.
- HSRP is a Cisco-proprietary FHRP designed to allow for transparent failover of a first-hop IP device
  - The virtual router represents an IP address that all routers in a HSRP group respond to. It serves as the default gateway setting of LAN hosts
  - The active router is a physical router that performs the actual task of forwarding packets
  - The standby router is backup physical router that will take over forwarding duties if the active router fails
- HSRP routers undergo an election process to determine their roles. The router with the highest priority becomes the active router; while the second highest becomes the standby router
- An HSRP router can have preemption enabled to allow it to force an election whenever it its priority is higher than the active or standby router in a group

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What did you learn in this module?

- HSRP routers undergo the following state changes:

  Init → Learn → Listen → Speak → Standby → Active

  Any additional routers will stabilize in the Listen state when an active and standby router are both present in a group

- Interface tracking allows routers to automatically lower their HSRP priority upon failure of a critical interface in order to relinquish an active or standby role

- When using STP and FHRP together in a network, alignment of the spanning tree root and the active router is crucial for efficient traffic flow

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